



ME 342A: SYSTEM DESIGN AND ANALYSIS

UNIT 3 SYSTEM OPTIMIZATION

Learning outcomes

- **Formulation of System Design Problem**
- **Application of Search Methods**
- **Linear Programming for Optimum Solutions**
- **Dynamic Programming for Optimum Solutions**



Linear programming (LP) is a central topic in optimization. It provides a powerful tool in modelling many applications. It is a mathematical technique to help plan and make decisions relative to the trade-offs necessary to allocate resources. It finds the minimum or maximum value of the objective and guarantees the optimal solution to the model formulated.

Applicability: There are many real-world applications that can be modelled as linear programming.

Solvability: There are theoretically and practically efficient techniques for solving large-scale problems.

Applications of LP:

1. Scheduling school buses to minimize total distance traveled.
2. Allocating police patrol units to high crime areas to minimize response time to calls.
3. Scheduling tellers at banks so that needs are met during each hour of the day while minimizing the total cost of labor.
4. Selecting the product mix in a factory to make best use of machine-and labor-hours available while maximizing the firm's profit.
5. Picking blends of raw materials in feed mills to produce finished feed combinations at minimum costs.
6. Determining the distribution system that will minimize total shipping cost.
7. Developing a production schedule that will satisfy future demands for a firm's product and at the same time minimize total production and inventory costs.
8. Allocating space for a tenant mix in a new shopping mall to maximize revenues to the leasing company

Requirements of an LP Problem:

- a) LP problems seek to maximize or minimize some quantity (usually profit or cost) expressed as an objective function.



- b) The presence of restrictions, or constraints, limits the degree to which we can pursue our objective.
- c) There must be alternative courses of action to choose from.
- d) The objective and constraints in linear programming problems must be expressed in terms of linear equations or inequalities.

Components of an LP:

1. Decision variables: describe our choices that are under our control. We often start with identifying decision variables (i.e., what we want to determine among those things which are under our control). In most problems, the decision variables are required to be nonnegative, and this should be explicitly included in the formulation.
2. Objective function: describes a criterion that we wish to minimize (e.g., cost) or maximize (e.g., profit). We usually seek a criterion (or a measure) to compare alternative solutions. This yields the objective function.
3. Constraints: describe the limitations that restrict our choices for decision variables. In many practical problems, there are limitations (such as resource / physical / strategic / economical) that restrict our decisions. We describe these limitations using mathematical constraints.

linear programming (LP) is known as an optimization problem in which the objective function is linear, and each constraint is a linear inequality or equality

Example 1

A company is involved in the production of two items (X and Y). The resources need to produce X and Y are twofold, namely machine time for automatic processing and craftsman time for hand finishing. The table below gives the number of minutes required for each item:

	Machine time	Craftsman time
Item X	13	20
Y	19	29



The company has 40 hours of machine time available in the next working week but only 35 hours of craftsman time. Machine time is costed at rupees 10 per hour worked and craftsman time is costed at rupees 2 per hour worked. Both machine and craftsman idle times incur no costs. The revenue received for each item produced (all production is sold) is rupees 20 for X and rupees 30 for Y. The company has a specific contract to produce 10 items of X per week for a customer. Formulate the problem of deciding how much to produce per week as a linear program.

Solution

Let

x be the number of items of X

y be the number of items of Y

then the LP is:

maximise

$$20x + 30y - 10(\text{machine time worked}) - 2(\text{craftsman time worked})$$

subject to:

$$13x + 19y \leq 40(60) \text{ machine time}$$

$$20x + 29y \leq 35(60) \text{ craftsman time}$$

$$x \geq 10 \text{ contract}$$

$$x, y \geq 0$$

so that the objective function becomes

maximise

$$20x + 30y - 10(13x + 19y)/60 - 2(20x + 29y)/60$$

i.e. maximise

$$17.1667x + 25.8667y$$

subject to:

$$13x + 19y \leq 2400$$

$$20x + 29y \leq 2100$$



$$x \geq 10$$

$$x, y \geq 0$$

Example 2

A company manufactures two products (A and B) and the profit per unit sold is rupees 3 and rupees 5, respectively. Each product must be assembled on a machine, each unit of product A taking 12 minutes of assembly time and each unit of product B 25 minutes of assembly time. The company estimates that the machine used for assembly has an effective working week of only 30 hours (due to maintenance/breakdown).

Technological constraints mean that for every five units of product A produced at least two units of product B must be produced. Formulate the problem of how much of each product to produce as a linear program.

The company has been offered the chance to hire an extra machine, thereby doubling the effective assembly time available. What is the maximum amount you would be prepared to pay (per week) for the hire of this machine and why?

Solution

Let

x_A = number of units of A produced

x_B = number of units of B produced

then the constraints are:

$$12x_A + 25x_B \leq 30(60) \text{ (assembly time)}$$

$$x_B \geq 2(x_A/5)$$

$$\text{i.e. } x_B - 0.4x_A \geq 0$$

$$\text{i.e. } 5x_B \geq 2x_A \text{ (technological)}$$

where $x_A, x_B \geq 0$

and the objective is

$$\text{maximise } 3x_A + 5x_B$$

Solving simultaneously, we have that:



$$x_A = (1800/22) = 81.8$$

$$x_B = 0.4x_A = 32.7$$

with the value of the objective function being rupees 408.9

Doubling the assembly time available means that the assembly time constraint (currently $12x_A + 25x_B \leq 1800$) becomes $12x_A + 25x_B \leq 2(1800)$. This new constraint will be parallel to the existing assembly time constraint so that the new optimal solution will lie at the intersection of $12x_A + 25x_B = 3600$ and $x_B - 0.4x_A = 0$

$$\text{i.e. at } x_A = (3600/22) = 163.6$$

$$x_B = 0.4x_A = 65.4$$

with the value of the objective function being rupees 817.8.

Hence, we have made an additional profit of rupees $(817.8 - 408.9) =$ rupees 408.9 and this is the maximum amount we would be prepared to pay for the hire of the machine for doubling the assembly time.

This is because if we pay more than this amount then we will reduce our maximum profit below the rupees 408.9 we would have made without the new machine.

Example 3

A company makes two products (P and Q) using two machines (A and B). Each unit of P that is produced requires 50 minutes processing time on machine A and 30 minutes processing time on machine B. Each unit of Q that is produced requires 24 minutes processing time on machine A and 33 minutes processing time on machine B. Machine A is going to be available for 40 hours and machine B is available for 35 hours. The profit per unit of P is Rs. 25 and the profit per unit of Q is Rs. 30. Formulate this problem to determine the production quantity of each product in such a way as to maximize the total profit given that the available resources should not be exceeded.

Solution



The objective here is to formulate the problem of deciding how much of each product to make in the current week as an LP.

As the company wants to determine the optimal product to make in the current week. So, there are two decision variables:

x: the number of units of P

y: the number of units of Q

We want to maximize the total profit. The profit per each unit of product P is Rs. 25 and profit per each unit of product Q is Rs. 30. Therefore, the total profit is $25x+30y$ if we produce x units of P and y units of Q. This leads to the following objective function:

$$\max 25x+30y$$

The objective function is linear in terms of decision variables x and y (i.e., it is of the form $ax + by$, where a and b are constant). We typically use the variable z to denote the value of the objective. So, the objective function can be stated as:

$$\max z=25x+30y$$

The amount of time that machine A is available restricts the quantities to be manufactured. If we produce x units of P and y units of Q, machine A should be used for $50x+24y$ minutes since each unit of P requires 50 minutes processing time on machine A and each unit of Q requires 24 minutes processing time on machine A. On the other hand, machine A is available for 40 hours or equivalently for 2400 minutes. This imposes the following constraint:

$$50x + 24y \leq 2400$$

Similarly, the amount of time that machine B is available imposes the following constraint:

$$30x + 33y \leq 2100.$$

These constraints are linear inequalities since in each constraint the left-hand side of the inequality sign is a linear function in terms of the decision variables x and



y and the right-hand side is constant. We need to include the following two nonnegativity constraints as well:

$$x \geq 0 \text{ and } y \geq 0$$

So, the constraints we are subject to (s.t.) are:

$$50x + 24y \leq 2400, \text{ (machine A time)}$$

$$30x + 33y \leq 2100, \text{ (machine B time)}$$

$$x \geq 0, y \geq 0.$$

Here is the LP:

$$\max z = 25x + 30y$$

$$\text{s.t. } 50x + 24y \leq 2400,$$

$$30x + 33y \leq 2100,$$

$$x \geq 0, y \geq 0.$$

Example 4

An operations manager is trying to determine a production plan for the next week. There are three products (P, Q, and R) to produce using four machines (A and B, C, and D). Each of the four machines performs a unique process. There is one machine of each type, and each machine is available for 2400 minutes per week. The unit processing times for each machine is given in Table 1.

Table 1: Machine Data

Unit Processing Time (min)				
Machine	Product P	Product Q	Product R	Availability (min)
A	20	10	10	2400
B	12	28	16	2400
C	15	6	16	2400
D	10	15	0	2400
Total processing time	57	59	42	9600



The unit revenues and maximum sales for the week are indicated in Table 2. Storage from one week to the next is not permitted. The operating expenses associated with the plant are Rs. 6000 per week, regardless of how many components and products are made. Rs. 6000 includes all expenses except for material costs.

Table 2: Product Data

Item	Product P	Product Q	Product R
Revenue per unit	Rs. 90	Rs. 100	Rs. 70
Material cost per unit	Rs. 45	Rs. 40	Rs. 20
Profit per unit	Rs. 45	Rs. 60	Rs. 50
Maximum sales	100	40	60

Objective is to find the optimal product mix that is, the amount of each product that should be manufactured during the present week to maximize profits. Formulate this as an LP.

Solution

Here we seek the “optimal” product mix, i.e., the amount of each product that should be manufactured during the present week to maximize profits. We are trying to select the optimal product mix, so we define three decision variables as follows:

p: number of units of product P to produce,

q: number of units of product Q to produce,

r: number of units of product R to produce.

Our objective is to maximize profit:

$$\begin{aligned}\text{Profit} &= (90-45)p + (100-40)q + (70-20)r - 6000 \\ &= 45p + 60q + 50r - 6000\end{aligned}$$

As the operating costs are not a function of the variables in the problem. If we were to drop the Rs. 6000 term from the profit function, we would still obtain the same optimal mix of products. Thus, the objective function is:



$$z = 45p + 60q + 50r$$

The amount of time a machine is available and the maximum sales potential for each product restrict the quantities to be manufactured. Since we know the unit processing times for each machine, the constraints can be written as linear inequalities as follows:

$$20p+10q+10r \leq 2400 \text{ (Machine A)}$$

$$12p+28q+16r \leq 2400 \text{ (Machine B)}$$

$$15p+6q+16r \leq 2400 \text{ (Machine C)}$$

$$10p+15q+0r \leq 2400 \text{ (Machine D)}$$

Observe that the unit for these constraints is minutes per week. Both sides of an inequality must be in the same unit. The market limitations are written as simple upper bounds. Market constraints:

$$P \leq 100, Q \leq 40, R \leq 60$$

Logic indicates that we should also include nonnegativity restrictions on the variables. So, the nonnegativity constraints are:

$$P \geq 0, Q \geq 0, R \geq 0.$$

By combining the objective function and the constraints, we obtain the LP model as follows:

$$\max z=45p+60q+50r$$

$$\text{s.t. } 20p+10q+10r \leq 2400$$

$$12p+28q+16r \leq 2400$$

$$15p+6q+16r \leq 2400$$

$$10p+15q+0r \leq 2400$$

$$0 \leq p \leq 100$$

$$0 \leq q \leq 40$$

$$0 \leq r \leq 60$$

The optimal solution to this LP is $P=81.82$, $Q=16.36$, $R=60$ with the corresponding objective value $z=\text{Rs. } 7664$. To compute the profit for the week,



we reduce this value by Rs. 6000 for operating expenses and get Rs. 1664. By setting the production quantities to the amount specified by the solution, we find machine usage as shown in the following table.

Machine data

Machine	Availability (min)	Actual use (min)
A	2400	2400
B	2400	2400
C	2400	2285
D	2400	1064

Product data

Item	Product P	Product Q	Product R
Maximum sales	100	40	60
Units produced	81.82	16.63	60

Example 5

Solve

minimise

$$4a + 5b + 6c$$

subject to

$$a + b \geq 11$$

$$a - b \leq 5$$

$$c - a - b = 0$$

$$7a \geq 35 - 12b$$

$$a \geq 0 \quad b \geq 0 \quad c \geq 0$$

Solution

To solve this LP we use the equation $c - a - b = 0$ to put $c = a + b$ (≥ 0 as $a \geq 0$ and $b \geq 0$) and so the LP is reduced to

minimise

$$4a + 5b + 6(a + b) = 10a + 11b$$

subject to

$$a + b \geq 11$$



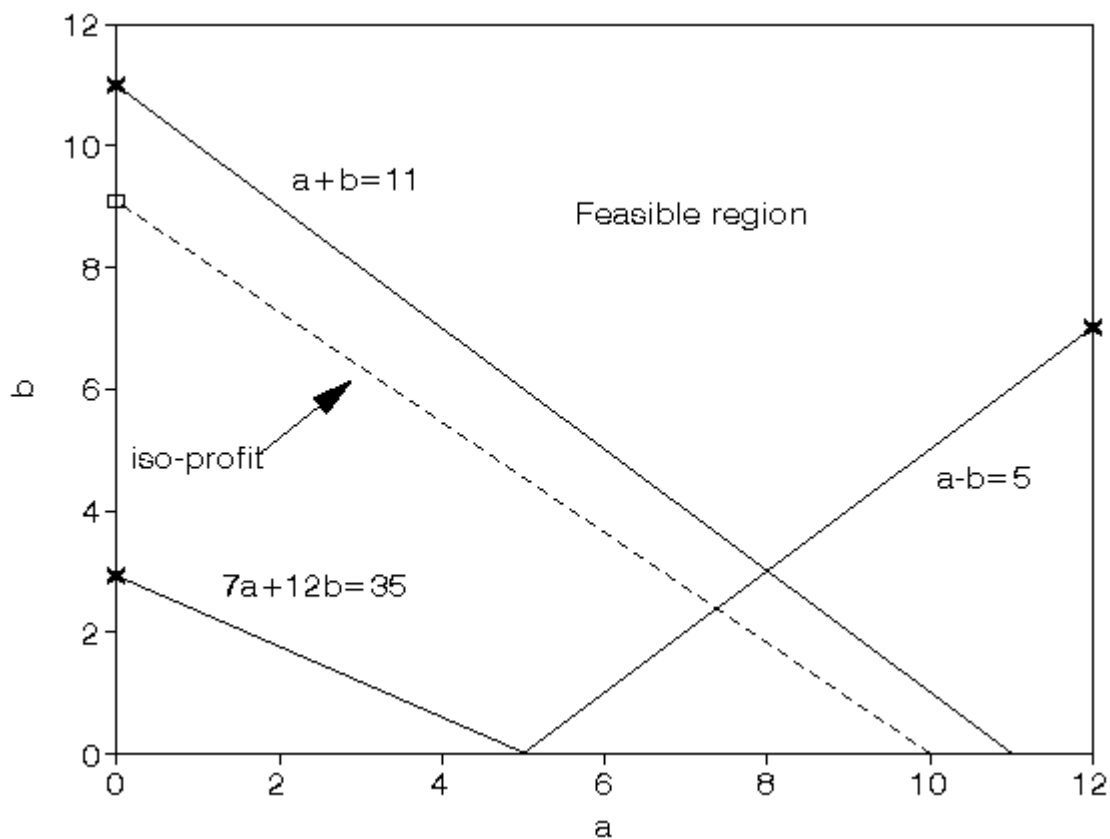
$$a - b \leq 5$$

$$7a + 12b \geq 35$$

$$a \geq 0 \quad b \geq 0$$

From the diagram below the minimum occurs at the intersection of $a - b = 5$ and $a + b = 11$

i.e. $a = 8$ and $b = 3$ with $c (= a + b) = 11$ and the value of the objective function $10a + 11b = 80 + 33 = 113$.



Example 6

Solve the following linear program:

$$\text{maximise } 5x_1 + 6x_2$$

subject to

$$x_1 + x_2 \leq 10$$

$$x_1 - x_2 \geq 3$$

$$5x_1 + 4x_2 \leq 35$$

$$x_1 \geq 0$$



$$x_2 \geq 0$$

Solution

It is plain from the diagram below that the maximum occurs at the intersection of

$$5x_1 + 4x_2 = 35 \text{ and}$$

$$x_1 - x_2 = 3$$

Solving simultaneously, rather than by reading values off the graph, we have that

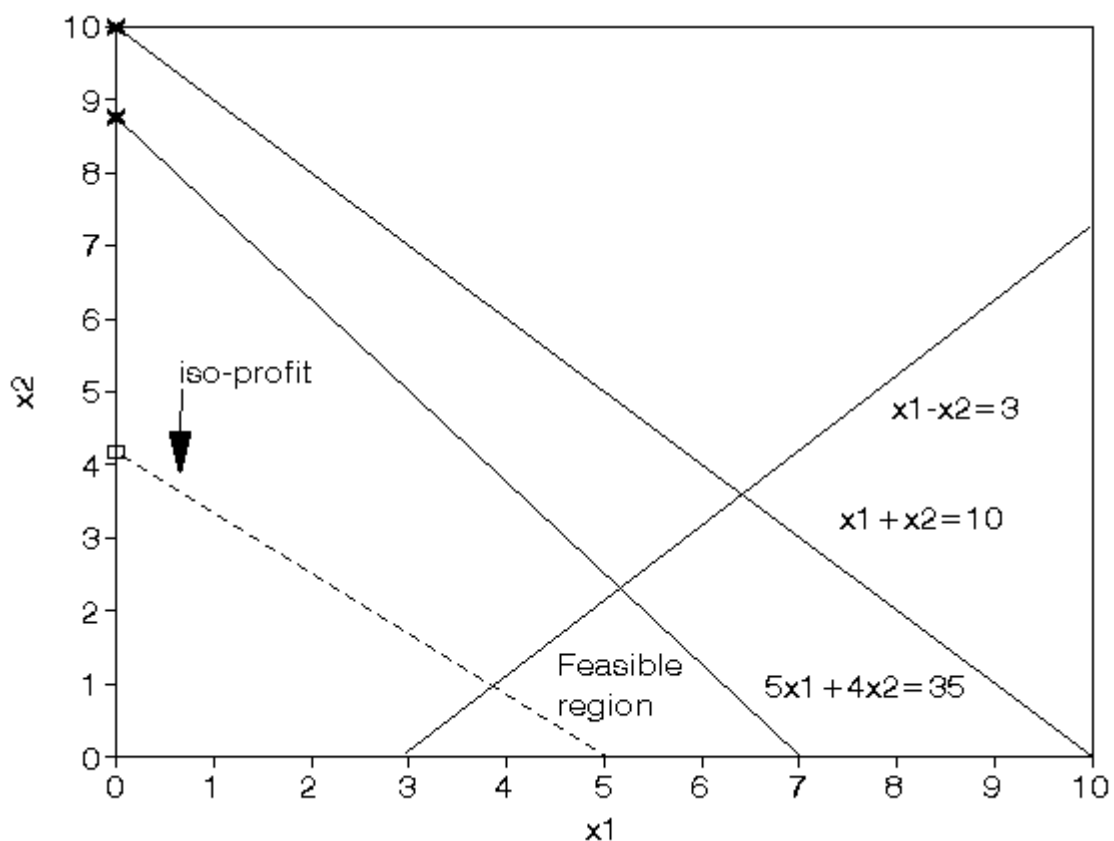
$$5(3 + x_2) + 4x_2 = 35$$

$$\text{i.e. } 15 + 9x_2 = 35$$

$$\text{i.e. } x_2 = (20/9) = 2.222 \text{ and}$$

$$x_1 = 3 + x_2 = (47/9) = 5.222$$

The maximum value is $5(47/9) + 6(20/9) = (355/9) = 39.444$



Example 7

A carpenter makes tables and chairs. Each table can be sold for a profit of rupees 30 and each chair for a profit of rupees 10. The carpenter can afford to spend up



to 40 hours per week working and takes six hours to make a table and three hours to make a chair. Customer demand requires that he makes at least three times as many chairs as tables. Tables take up four times as much storage space as chairs and there is room for at most four tables each week.

Formulate this problem as a linear programming problem and solve it graphically.

Solution

Variables

Let

x_T = number of tables made per week

x_C = number of chairs made per week

Constraints

total work time

$$6x_T + 3x_C \leq 40$$

customer demand

$$x_C \geq 3x_T$$

storage space

$$(x_C/4) + x_T \leq 4$$

all variables ≥ 0

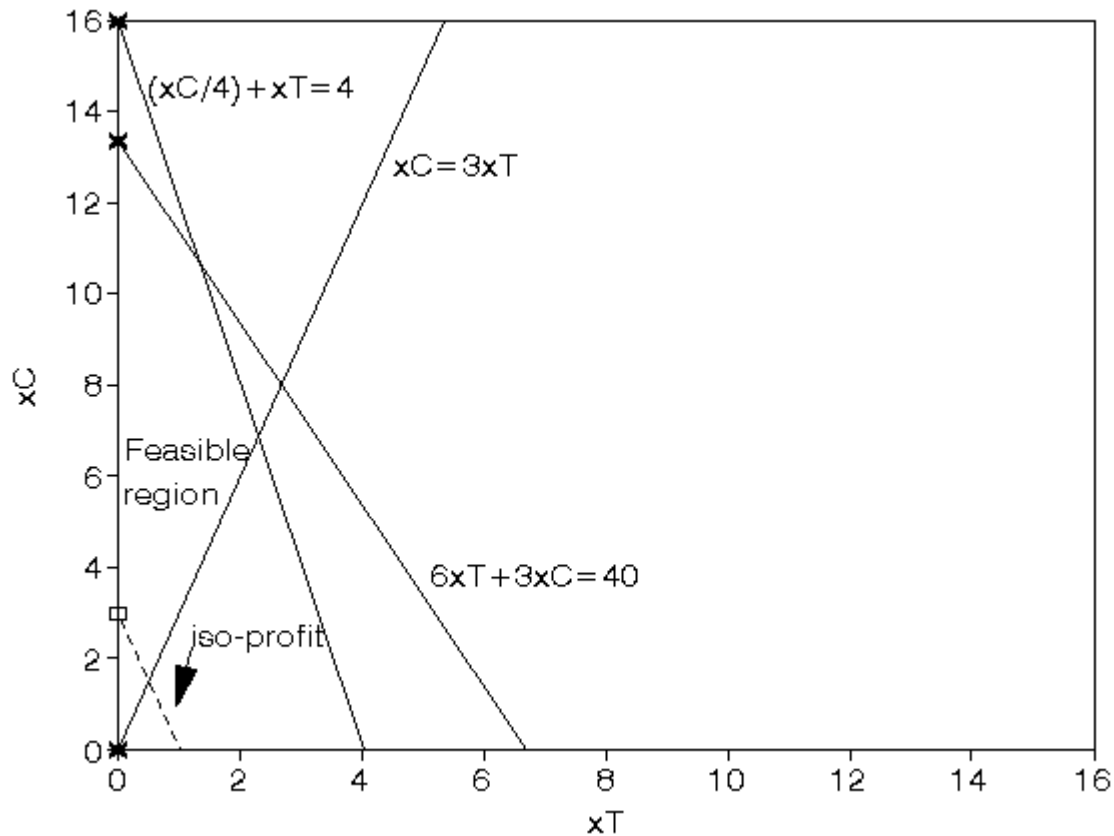
Objective

maximise $30x_T + 10x_C$

The graphical representation of the problem is given below and from that we have that the solution lies at the intersection of

$$(x_C/4) + x_T = 4 \text{ and } 6x_T + 3x_C = 40$$

Solving these two equations simultaneously we get $x_C = 10.667$, $x_T = 1.333$ and the corresponding profit = Rs. 146.667



Dynamic Programming

Dynamic programming (DP) is a methodology applied primarily to sequential decision processes, such as those occurring in stages over time. As, DP methods are well-suited for problems with a time dimension, they arise naturally in financial settings. The term dynamic programming refers both to a modelling paradigm and to a specific set of methodologies. DP methods are based on Bellman's Principle of Optimality. According to this principle, in a sequential decision process, every subsequence of an optimal decision sequence must also be optimal when viewed as a separate decision problem. This principle enables us to formulate recursive relationships that lead to algorithms for solving optimization problems.

The common elements of a DP include

1. A set of decision stages
2. A set of possible states in each decision stage



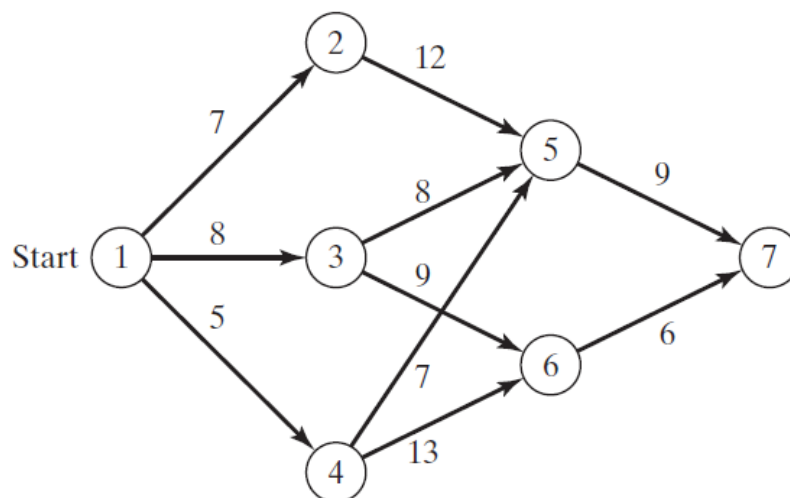
3. A set of transitions between states
4. A value function for each state in each stage indicating the best objective value achievable from that state.

Each transition has a cost and can be associated with either an action by the decision-maker or a random event. There are implicit constraints that determine what transitions are feasible.

The main idea of DP is to decompose the problem into (more manageable) subproblems. Computations are then carried out recursively where the optimum solution of one subproblem is used as an input to the next subproblem. The optimum solution for the entire problem is at hand when the last subproblem is solved. The way the recursive computations are carried out depends on how the original problem is decomposed. In particular, the subproblems are normally linked by common constraints. The feasibility of these common constraints is maintained at all iterations.

Example

Solve the problem of the shortest highway route between two cities by Dynamic Programming. The network in below figure provides the possible routes between the starting city at node 1 and the destination city at node 7. The routes pass through intermediate cities designated by nodes 2 to 6.





Solution:

We can solve this problem by enumerating all the routes between nodes 1 and 7 (there are five such routes). However, exhaustive enumeration is computationally intractable in large networks. To solve the problem by DP, first decompose it into *stages*. Next, carry out the computations for each stage separately.

The general idea for determining the shortest route is to compute the shortest (cumulative) distances to all the terminal nodes of a stage and then use these distances as input data to the immediately succeeding stage. Starting from node 1, stage 1 reaches three end nodes (2, 3, and 4), and its computations are simple.

Stage 1 Summary

Shortest distance from node 1 to node 2 = 7 miles (*from node 1*)

Shortest distance from node 1 to node 3 = 8 miles (*from node 1*)

Shortest distance from node 1 to node 4 = 5 miles (*from node 1*)

Next, stage 2 has two end nodes, 5 and 6. Node 5 can be reached from nodes 2, 3, and 4 via routes (2, 5), (3, 5), and (4, 5). This information, together with the summary results (shortest distances) in stage 1, determines the shortest (cumulative) distance to node 5 as

$$\begin{aligned}
 & \text{(Shortest distance to node 5)} \\
 &= \min_{i=2,3,4} \{ \text{(Shortest distance to node } i) \\
 &\quad + \text{(Distance from node } i \text{ to node 5)} \} \\
 &= \min \left\{ \begin{array}{l} 7 + 12 = 19 \\ 8 + 8 = 16 \\ 5 + 7 = 12 \end{array} \right\} = 12 \text{(from node 4)}
 \end{aligned}$$

Node 6 can be reached from nodes 3 and 4 only. Thus

$$\begin{aligned}
 & \text{(Shortest distance to node 6)} \\
 &= \min_{i=3,4} \{ \text{(Shortest distance to node } i) \\
 &\quad + \text{(Distance from node } i \text{ to node 6)} \} \\
 &= \min \left\{ \begin{array}{l} 8 + 9 = 17 \\ 5 + 13 = 18 \end{array} \right\} = 17 \text{(from node 3)}
 \end{aligned}$$



Stage 2 Summary

Shortest distance from node 1 to node 5 = 12 miles (from node 4)

Shortest distance from node 1 to node 6 = 17 miles (from node 3)

The last step is to consider stage 3. The destination node 7 can be reached from either node 5 or 6. Using the summary results from stage 2 and the distances from nodes 5 and 6 to node 7, we get

$$\begin{aligned}
 & (\text{Shortest distance to node 7}) \\
 &= \min_{i=5,6} \{ (\text{Shortest distance to node } i) \\
 &\quad + (\text{Distance from node } i \text{ to node 7}) \} \\
 &= \min \left\{ \begin{array}{l} 12 + 9 = 21 \\ 17 + 6 = 23 \end{array} \right\} = 21 (\text{from node 5})
 \end{aligned}$$

Stage 3 Summary

Shortest distance from node 1 to node 7 = 21 miles (from node 5)

Stage 3 summary shows that the shortest distance between nodes 1 and 7 is 21 miles. To determine the optimal route, start at stage 3 summary, where node 7 links to node 5; stage 2 summary links node 4 to node 5; and stage 1 summary links node 4 to node 1. Thus, the shortest route is 1-4-5-7.

The example reveals the basic properties of DP computations:

1. The computations at each stage are a function of the feasible routes of that stage, and only that stage.
2. A current stage is linked to the *immediately preceding* stage only (without regard to earlier stages) based on the shortest-distance summary of the immediately preceding stage.

Principle of optimality

Future decisions for all *future* stages constitute an optimal policy regardless of the policy adopted in all *preceding* stages. The implementation of the principle of optimality is evident in the computations in above example. In stage 3, the recursive computations at node 7 use the shortest distance to nodes 5 and 6 (i.e.,



the states of stage 2) without concern about how nodes 5 and 6 are reached from the starting node 1.

The principle of optimality does not address the details of how a subproblem is optimized. The reason is the generic nature of the subproblem. It can be linear or nonlinear, and the number of alternatives can be finite or infinite. All the principle of optimality does is “break down” the original problem into more computationally tractable subproblems.

Forward and Backward Recursion

Above example used **forward recursion** in which the computations proceed from stage 1 to stage 3. The same example can be solved by **backward recursion**, starting at stage 3 and ending at stage 1. Naturally, both the forward and backward recursions yield the same optimum. Although the forward procedure appears more logical, DP literature mostly uses backward recursion. The reason for this preference is that, in general, backward recursion can be more efficient computationally.

Now, the use of backward recursion is demonstrated by applying it to above example. The demonstration will also provide the opportunity to present the DP computations in a compact tabular form.

Stage 3

Node 7 ($x_4 = 7$) is connected to nodes 5 and 6 ($x_3 = 5$ and 6) with exactly one route each. The following table summarizes stage 3 computations:

x_3	$d(x_3, x_4)$	Optimum solution	
	$x_4 = 7$	$f_3(x_3)$	x_4^*
5	9	9	7
6	6	6	7

Stage 2

Route (2, 6) does not exist. Given $f_3(x_3)$ from stage 3, we can compare the feasible alternatives as the following table shows:



x_2	$d(x_2, x_3) + f_3(x_3)$		Optimum solution	
	$x_3 = 5$	$x_3 = 6$	$f_2(x_2)$	x_3^*
2	$12 + 9 = 21$	--	21	5
3	$8 + 9 = 17$	$9 + 6 = 15$	15	6
4	$7 + 9 = 16$	$13 + 6 = 19$	16	5

The optimum solution of stage 2 reads as follows:

For cities 2 and 4, the shortest routes pass through city 5, and for city 3, the shortest route passes through city 6.

Stage 1

From node 1, we have three alternative routes: (1, 2), (1, 3), and (1, 4). Using $f_2(x_2)$ from stage 2, we get

x_1	$d(x_1, x_2) + f_2(x_2)$			Optimum solution	
	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$	$f_1(x_1)$	x_2^*
1	$7 + 21 = 28$	$8 + 15 = 23$	$5 + 16 = 21$	21	4

Stage 1 solution links city 1 to city 4. Next, stage 2 solution links city 4 to city 5. Finally, stage 3 solution connects city 5 to city 7. The optimum route is 1-4-5-7, and the associated distance is 21 miles.